

# NETFRAME

## Homelab Infrastructure Runbook

km-cluster | NetFRAME CS9000 42U

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**Revised:** 2026-07-10  
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## 1 Infrastructure Overview

NetFRAME is a professional-grade homelab built inside a NetFRAME CS9000 42U rack, purpose-built as a live CCNA lab, LLM inference platform, ML research node, and backup infrastructure. The cluster is named **km-cluster** and runs Proxmox VE 9.2.3 across seven nodes as of July 2026 (Randy stayed on 9.1.1 — a kernel/ZFS-only upgrade).

### 1.1 Cluster Nodes

| Hostname  | Role          | IP             | CPU            | RAM    | Notes                    |
|-----------|---------------|----------------|----------------|--------|--------------------------|
| Randy     | Storage / PBS | 192.168.10.187 | 2x E5-2690 v3  | 128 GB | SuperMicro CSE-219U      |
| QuarkyLab | ML / research | 192.168.10.179 | 2x E5-2699 v4  | 512 GB | R730, RTX 8000 48GB      |
| Jarvis    | LLM Inference | 192.168.10.31  | 2x E5-2687W v4 | 384 GB | R730, 2x RTX 6000 (48GB) |
| pve2      | OPNsense host | 192.168.10.204 | i7-8700        | 32 GB  | HP EliteDesk G4 SFF      |
| pve3      | Cluster node  | 192.168.10.201 | i7-8700        | 48 GB  | HP EliteDesk G4 SFF      |
| pve4      | Cluster node  | 192.168.10.202 | i5-7500T       | 32 GB  | HP EliteDesk G3 Mini     |
| pve5      | Cluster node  | 192.168.10.203 | i5-7500T       | 32 GB  | HP EliteDesk G3 Mini     |

### 1.2 Management IPs

| Device          | IP                                                          |
|-----------------|-------------------------------------------------------------|
| EX3400 (switch) | 192.168.10.50                                               |
| OPNsense LAN    | 192.168.10.1                                                |
| Randy IPMI      | 192.168.20.22 (VLAN 20)                                     |
| QuarkyLab iDRAC | 192.168.20.20 (VLAN 20)                                     |
| Jarvis iDRAC    | 192.168.20.21 (VLAN 20)                                     |
| Pi-hole / DNS   | 192.168.10.177                                              |
| Ares (admin)    | ares.netframe.local = 192.168.10.152 (Wi-Fi) / .131 (wired) |

## 2 Network Architecture

### 2.1 VLAN Design

| VLAN | Name    | Subnet          | Notes                   |
|------|---------|-----------------|-------------------------|
| 1    | mgmt    | 192.168.10.0/24 | Management, all servers |
| 20   | trusted | 192.168.20.0/24 | Trusted devices         |
| 30   | servers | 192.168.30.0/24 | Service VMs             |
| 40   | IoT     | 192.168.40.0/24 | IoT devices             |
| 50   | VoIP    | 192.168.50.0/24 | Cisco 8841 phones       |
| 60   | guest   | 192.168.60.0/24 | Guest WiFi              |
| 70   | lab     | 192.168.70.0/24 | CCNA lab segment        |

### 2.2 Switching

- **Juniper EX3400-48P** — enterprise fabric, JunOS 23.4R2-S7.4, IP 192.168.10.50

- **UniFi Switch 24 PRO (PoE+)** — consumer fabric (IoT, VoIP, guest)
- Inter-switch uplink: ge-0/0/32 (copper RJ45, active); xe-0/2/3 DAC (link-down, speed mismatch)

## 2.3 10G Fabric — ConnectX-3 DAC Links

| Node              | NIC Port   | EX3400 Port                            |
|-------------------|------------|----------------------------------------|
| Randy (nic3)      | ConnectX-3 | xe-0/2/0 — Up 10G (VLAN 30)            |
| Jarvis (enp132s0) | ConnectX-3 | xe-0/2/2 — Up 10G (VLAN 30)            |
| QuarkyLab         | onboard 1G | ge-0/0/24 — trunk (native 1 + VLAN 30) |

## 2.4 OPNsense

OPNsense runs as VM 100 on pve2 (192.168.10.204). Version 25.1.12. Handles inter-VLAN routing, firewall, DHCP for all VLANs, and WAN. Serial console access: `qm terminal 100` from pve2.

## 3 Randy — SuperMicro CSE-219U

### Randy — Commissioned 2026-06-22

**Role:** Primary storage node, Proxmox Backup Server  
**IP:** 192.168.10.187    **IPMI:** 192.168.20.22 (VLAN 20)  
**OS:** Proxmox VE 9.1.1 (kernel 7.0.12-1-pve)  
**CPU:** Dual Intel E5-2690 v3 (24 cores / 48 threads)  
**RAM:** 128 GB ECC DDR4

## 3.1 Storage Layout

| Device                  | Type                | Purpose                      |
|-------------------------|---------------------|------------------------------|
| /dev/sda (RAID-1)       | 185.8 GB SAS SSD x2 | Proxmox OS boot mirror       |
| 18x Toshiba AL15SEB18EQ | 1.636TB 10K SAS     | datastore — 3x 6-wide RAIDZ2 |
| 4x Seagate ST2000NX0423 | 1.819TB SATA        | datastore — 1x 4-wide RAIDZ2 |

### 3.1.1 RAID Controller — AVAGO 3108 MegaRAID

- Boot mirror: RAID-1 on slots 0:0 and 0:1 (Seagate ST200FM0053 SSDs)
- Data drives: JBOD mode enabled, all 22 data drives passed through to OS
- StorCLI installed at: `/opt/MegaRAID/storcli/storcli64`
- Symlink: `/usr/sbin/storcli64`

Listing 1: Useful StorCLI commands

```
# Show all drives and state
storcli64 /c0/eall/sall show

# Show controller info
```

```
storcli64 /c0 show

# Enable JBOD mode (if needed after reboot)
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod
```

### 3.1.2 ZFS Pool — datastore

Listing 2: ZFS pool layout

```
zpool status datastore
# pool: datastore
# state: ONLINE
#   raidz2-0: 6x Toshiba 1.636TB
#   raidz2-1: 6x Toshiba 1.636TB
#   raidz2-2: 6x Toshiba 1.636TB
#   raidz2-3: 4x Seagate ST2000NX0423 1.819TB
# 36.7TB raw / ~23TB usable (all drives in-pool; no spares)
```

## 3.2 Proxmox Backup Server

- PBS version: 4.2.2
- Web UI: <https://192.168.10.187:8007>
- Datastore: `datastore` at `/datastore`
- Fingerprint: stored in Vaultwarden (not published)
- Storage ID in cluster: `randy-pbs` (Shared: Yes, all nodes)

## 3.3 Networking

- **nic3** (Mellanox ConnectX-3): 10G UP → xe-0/2/0 on EX3400
- **Dual-homed (2026-07-02)**: mgmt/corosync on VLAN 1 (192.168.10.187); NFS/PBS/egress on VLAN 30 (192.168.30.187)
- **nic0, nic1** (onboard Intel): unused
- **vmbr0** bridges **nic3** for management

### 3.3.1 Randy Commissioning Procedure (for rebuild reference)

Listing 3: Full Randy rebuild procedure

```
# 1. Boot IPMI BIOS -- hit Ctrl+H for MegaRAID WebBIOS
#   Create RAID-1 VD on two 185GB Seagate SSDs (slots 0:0, 0:1)

# 2. Install Proxmox VE 9.1 via USB
#   Target: /dev/sda (the RAID-1 VD, 185.78GiB, SMC3108)
#   Filesystem: ext4
#   Hostname: randy.netframe.local
#   IP: 192.168.10.187/24   GW: 192.168.10.1   DNS: 192.168.10.177
```

```
# 3. Post-install repo fix
echo "# disabled" > /etc/apt/sources.list.d/pve-enterprise.list
echo "# disabled" > /etc/apt/sources.list.d/ceph.list
echo "" > /etc/apt/sources.list.d/pve-enterprise.sources
echo "" > /etc/apt/sources.list.d/ceph.sources
echo "deb http://download.proxmox.com/debian/pve trixie pve-no-
subscription" \
  > /etc/apt/sources.list.d/pve-no-subscription.list # trixie (Debian
13/PVE 9); NOT bookworm
apt update && apt dist-upgrade -y

# 4. Join cluster
echo "192.168.10.204 pve2.netframe.local pve2" >> /etc/hosts
pvecm add 192.168.10.204 --use_ssh

# 5. Enable JBOD passthrough
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod

# 6. Create ZFS pool
zpool create -f -o ashift=12 datastore \
  raidz2 sdb sdd sde sdf sdh sdi \
  raidz2 sdj sdk sdl sdm sdn sdo \
  raidz2 sdp sdq sdr sds sdt sdu \
  raidz2 <4x Seagate ST2000NX0423> # 4th vdev, added in the 2026-06-25
expansion

# 7. Install PBS
echo "deb http://download.proxmox.com/debian/pbs trixie pbs-no-
subscription" \
  > /etc/apt/sources.list.d/pbs-no-subscription.list
apt update && apt install proxmox-backup-server -y

# 8. Create PBS datastore
proxmox-backup-manager datastore create datastore /datastore
```

## 4 QuarkyLab — Dell R730 #1

### QuarkyLab

**Role:** the researcher's ML research node

**iDRAC:** 192.168.20.20 (VLAN 20; creds in Vaultwarden)

**CPU:** Dual E5-2699 v4 (matched stepping) **RAM:** 512 GB LRDIMM ECC

**GPU:** RTX 8000 48 GB (installed 2026-07-01) **BIOS:** 2.19.0

**GPU stack (verified 2026-07-01):** NVIDIA 550.163.01 on kernel 6.14.11-9-pve, pinned via GRUB\_DEFAULT (6.17+ breaks the nvidia-550 DRM API). nvidia-smi reports 48 GB (driver-free Turing swap from the earlier RTX 6000).

Listing 4: Kernel pin for the NVIDIA 550 stack (QuarkyLab)

```
apt install pve-kernel-6.14.11-9-pve
```

```
# Pin as default in /etc/default/grub (GRUB_DEFAULT), then:
update-grub && reboot
# NVIDIA 550.163.01 driver already installed & verified -- do NOT
  upgrade the kernel
```

## 5 Jarvis — Dell R730 #2

### Jarvis

**Role:** LLM inference node (Ollama, Qwen2.5 72B)  
**iDRAC:** 192.168.20.21 (VLAN 20; creds in Vaultwarden)  
**CPU:** Dual E5-2687W v4    **RAM:** 384 GB LRDIMM ECC  
**GPU:** 2x RTX 6000 — 24GB each / 48GB total (installed 2026-07-04)

Ollama serves Qwen2.5 72B Q4\_K\_M, GPU-backed across both cards (models on the `tank/models` ZFS dataset). FastAPI `llm_router.py` implements hybrid routing — local Ollama by default, escalating to the Claude API on an explicit `escalate` flag, a `model=claude-*` request, or local failure (Ollama exposes no logprobs, so routing is by flag/model/failure, not confidence).

## 6 Services

### 6.1 Running Services

| Service   | Host      | VM/CT ID | IP                  | Notes                      |
|-----------|-----------|----------|---------------------|----------------------------|
| OPNsense  | pve2      | VM 100   | 192.168.10.1        | Router/firewall            |
| Pi-hole   | pve1      | LXC 103  | 192.168.10.177      | DNS (Mac Mini, standalone) |
| Headscale | pve5      | LXC 105  | 192.168.10.186      | VPN, v0.29.1               |
| Wazuh     | QuarkyLab | VM 104   | 192.168.10.184      | SIEM                       |
| step-ca   | pve2      | —        | —                   | Internal CA                |
| PBS       | Randy     | —        | 192.168.10.187:8007 | Backup server              |

### 6.2 Headscale

Headscale v0.29.1 running as LXC 105 on pve5. Phase 1 (2026-06-22) migrated Ares (100.64.0.1), Randy (.2), pve5 (.3), pve4 (.4), pve3 (.5) and Jarvis (.6) to self-hosted. Phase 2 (QuarkyLab + the researcher's Mac) is pending.

### 6.3 Internal CA (step-ca)

step-ca 0.30.2 on pve2 issuing `*.netframe.local` TLS certificates.

## 7 Power Infrastructure

| UPS                         | Feeds                        | Battery                          |
|-----------------------------|------------------------------|----------------------------------|
| Middle Atlantic UPS-OL2200R | R730s, Randy, DS4246         | 6x 12V 9Ah AGM in series (76.4V) |
| Tripp Lite SMART1500VA      | EX3400, UniFi, small compute | Stock                            |

PDU: APC AP7901 (ge-0/0/38 on EX3400).

## 8 Pending / To-Do

- ✓ VLAN activation — EX3400 trunk live, verified end-to-end (2026-06-25)
- ✓ QuarkyLab RTX 8000 + Jarvis 2x RTX 6000 installed & GPU-backed (2026-07)
- ✓ Backup schedules to randy-pbs (daily, 7d+4w retention)
- ✓ DS4246 — pool **bulk** built & online (2x 8-wide RAIDZ2, 41.3 TiB, 2026-07-08)
- ✓ Jellyfin on Randy (v10.11.11); RX 580 ROCm transcoding pending power cable
- ✓ UPS monitoring live (NUT → PeaNUT → Homepage/Grafana)
- Headscale Phase 2 — QuarkyLab + the researcher's Mac
- FreePBX VoIP (5x Cisco CP-8841 phones acquired)
- Cyberpunk monitoring dashboard — live API integration
- RKE2 / Kubernetes (Cilium, MetalLB, NVIDIA GPU Operator)

## 9 Quick Reference Commands

### 9.1 Cluster Health

```
# From any node
pvecm status
pvecm nodes

# PBS status
systemctl status proxmox-backup
proxmox-backup-manager datastore list

# ZFS pool health
zpool status datastore
zpool list
```

### 9.2 EX3400 Switch

```
ssh mason@192.168.10.50
# Always exit to > operational mode before show commands
# Paste one line at a time

show interfaces xe-0/2/0 brief      # Randy 10G
show interfaces xe-0/2/2 brief      # Jarvis 10G (ConnectX)
show ethernet-switching table      # MAC table
show lldp neighbors                 # Connected devices
```

### 9.3 Randy Storage

```
ssh root@192.168.10.187

# Check all drives
storcli64 /c0/eall/sall show

# ZFS status
zpool status datastore
zfs list

# If drives go dark after reboot (JBOD mode reset)
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod
```